Tenta 2020, frågor och svar

Vetenskapsteori och vetenskaplig metodik (naturvetenskap) (Kungliga Tekniska Högskolan)



Scan to open on Studocu

Studocu is not sponsored or endorsed by any college or university
Downloaded by jonathan X (jonathantades@gmail.com)

Final exam
Theory and Methodology of Science

Natural science

Course codes
AK2030/AK2036/AK2050/FAK3024/FAK3137/F1N5113

2020-03-09

General information

The exam consists of three parts corresponding to the learning outcomes of the course. Pen/pencil, rubber and dictionary (published paper dictionary, such as Swedish-English) are the only allowed external materials. Please do not use correction fluid (such as TipEx) since it makes the scanning process complicated, instead draw a line through any words you want to change.

The blue front page should be filled out with a black or blue pen, pencil is not allowed on this page. Please write clearly! The personal number should correspond to the one on the invigilator's list. You do not need to fill in the section "Mark attempted questions with X" at the bottom of the front page.

Grading: For each passing grade, you need to receive a minimum score on each of the parts as well as a minimum total score. You will find the grade thresholds in the tables below. For this exam, only bonus points from period 2 are valid. You could have received a maximum of 9.5 bonus points. Bonus points are added at the threshold of 20 points on the exam. The exam grade is then determined by your total score.

The exam is four hours long. You are free to leave after one hour. If you have questions regarding the content of the exam, please contact the invigilator.

Examiner: Prof. Till Grüne-Yanoff, Division of Philosophy, KTH. Visiting address: Teknikringen 76, bottom floor.

Downloaded by jonathan X (jonathantades@gmail.com)

Grade Tables

Bonus points are added at a score of 20 p, if and only if the minimum requirements on part I, part II, and part III are fulfilled. Note that *each* part's minimum requirement in the leftmost column must be fulfilled before the total points are converted into a grade.

Master students

Minimum requirements	Total score	Grade
PI: 8 PII: 6 PIII: 6	≥36.0 32–35.5 28–31.5 24–27.5 20–23.5 16–19.5 <16	A B C D E FX F
PI: 8 PII: 5 PIII: 5	≥32 28–31.5 24–27.5 20–23.5 16–19.5 <16	B C D E FX F
PI: 8 PII: 4 PIII: 4	≥28 24–27.5 20–23.5 16–19.5 <16	C D E FX F
PI: 8 PII: 3 PIII: 3	≥24 20–23.5 16–19.5 <16	D E FX F
PI: 8 PII: 2 PIII: 2	≥20 16–19.5 <16	E FX F
PI: 8 PII: 1 PIII: 1	≥16 <16	FX F
Otherwise...		F

PhD students

Minimum requirements	Total score	Grade
PI: 8 PII: 3 PIII: 3	≥28 <28	P F
Otherwise...		F

Downloaded by jonathan X (jonathantades@gmail.com)

PART I – Multiple choice questions

Instructions for multiple choice questions: For each question there is *at least* one correct answer alternative. In other words, for each question the number of correct alternatives is 1, 2, 3, or 4. Mark your answers with "X" on the separate answer sheet, *not* on the exam. You are awarded one point for each question if, and only if, you have selected the correct alternatives and no other. Partial points are not awarded on this part.

1. What is true about the justification of method choice?

   a) Choosing a method because it is suggested by teachers or peers is justification by convention.
   b) Science should primarily choose those methods that best fulfil epistemic and ethical goals.
   c) Scientists are sometimes not justified in their method choices.
   d) The practice of justifying method choice is called methodology.

2. What is true about *lexical* definitions? Select all true claims.

   a) A lexical definition attempts to capture how the author intends to use the term.
   b) A lexical definition attempts to capture how a dictionary would define a term.
   c) A lexical definition aims to capture the common usage of a term.
   d) A lexical definition can be imprecise.

3. What is true about *inductive inferences*? Select all true claims. Inductive inferences ...

   a) ... rerrange information contained in the premises, and do not amplify the information.
   b) ... require justification.
   c) ... have premises.
   d) ... never lead to true conclusions.

4. What is an *ad hoc* hypothesis? Select all correct alternatives.

   a) It is a modification of a theory that does not explain an observed anomaly.
   b) It is a modification of a theory that logically contradicts the rest of the theory.
   c) It is a modification of a theory, and the modification does not propose new means of testing the theory.
   d) It has fewer observable consequences than the previous hypothesis.

5. What is true about *falsifiable statements*? Select all true claims. A falsifiable statement...

   a) ... is either true or false.
   b) ... may be true.
   c) ... is necessarily false.
   d) ... may be false.

Downloaded by jonathan X (jonathantades@gmail.com)

6. What is true regarding theory-dependence?

a) Indirect observation requires theoretical justification of the connection between what is actually observed and the feature of interest.
b) Aided observations are not theory-dependent.
c) Some theories are required for making observations. These observations, in turn, justify these theories — this is called the experimenter's regress.
d) Since scientists gather data, from which they infer theory, basic research is thus not theory-dependent.

7. When one *ranks* movies on the scale 1,2,3,4,5, what kind of measurement scale is employed? Select all correct alternatives.

a) An interval scale.
b) A scale that represents less information than an interval scale.
c) An ordinal scale.
d) A scale that represents less information than a ratio scale.

8. What is correct about blinding in experiments? Select all correct alternatives.

a) You could use double blinding to reduce observer expectation effects (i.e. an interpretation problem).
b) Blinding can only be realized through randomization.
c) One condition for double blinding is that experimenters do not know which treatments are administered to what subjects.
d) Double blinding requires that the experimenter is ignorant of which variables are dependent and which are independent.

9. What is true about Randomised Control Trials (RCTs)?

a) Subjects are separated into test and control groups by a random process.
b) It guarantees that background factors are equally distributed between test and control in a test.
c) Using RCT implies that you are using double blinding.
d) Randomization helps implement single and double blinding.

10. What distinguishes models from theories? Select all correct claims.

a) Scientists who use an accepted theory have reasons to believe that it is true, while scientists who use an accepted model typically know that it is false in several respects.
b) By proposing a theory, one claims that the theory's target is indeed governed by the principles expressed in the theory. When using a model, one is typically aware of alternative representations that are useful for other purposes.
c) Theories make general claims, models can only represent particulars.
d) Theories allow for making true claims, models do not.

Downloaded by jonathan X (jonathantades@gmail.com)

11. What is true regarding the virtues of models?

    a) Transparency means that a model follows some scientific theory.
    b) Theoretical tractability means that a model can be processed by a theory, in a scientifically useful way.
    c) When comparing two models, the one which implies the parameter specifications of the other, is more precise.
    d) When a model user can understand how the model produces the result, then it is epistemically transparent.

12. Which claims are true about Bayesian statistics? Select all correct alternatives.

    a) Bayesian statistics interprets probabilities as "degrees of belief" or "subjective confidence in hypotheses".
    b) The main goal in Bayesian hypothesis evaluation is to decide whether to accept a single hypothesis as either true or false.
    c) In Bayesian statistics, the probability of a hypothesis is changed as a consequence of observing new evidence.
    d) In Bayesian statistics, no probabilities are assigned to hypotheses.

13. Why should hypotheses be tested statistically? Select all correct answers.

    a) Using probability calculus is a way to model and calculate the degree of confirmation that some evidence provides for a hypothesis.
    b) Statistics helps us specify an acceptable error margin for accepting or rejecting a hypothesis.
    c) Statistics is the only way to justify inductive inferences.
    d) Statistical accounts typically impose weaker (and hence often more adequate) demands on hypothesis testing than (non-statistical) accounts of confirmation and falsification.

14. What is true regarding the structure of explanations?

    a) An explanation features an explanandum, something to be explained, and an explanans, something which provides the explanation.
    b) An explanation features an explanans, something to be explained, and an explanandum, something which provides the explanation.
    c) Singular phenomena can only be given singular explanations.
    d) A singular explanation explains a particular event with reference to the relevant aspects of the particular situation.

15. Which claims are true about the relation between correlation and causation? Select all correct claims.

    a) Determining causal relations through Mill's method of difference is no longer a part of the modern practice of randomized controlled trials.
    b) Correlation consists in the association between variables. Causation consists in the productive influence of one variable on the other.
    c) That two variables are uncorrelated means that there is no causal relation between them.
    d) Correlation between two variables is symmetric. The causal relation between two variables is asymmetric.

Downloaded by jonathan X (jonathantades@gmail.com)

16. What is true of tacit knowledge?

a) The ability to ride a bicycle is an example of tacit knowledge.
b) Tacit knowledge is taught mainly through engineering textbooks and manuals.
c) Being able to describe the Navier-Stokes equations in detail is an example of tacit knowledge.
d) Tacit knowledge is difficult to transfer to another person by means of writing it down or verbalizing it.

17. What is true about belief-desire explanations? Select all correct alternatives.
Belief-desire explanations ...

a) ... are not causal explanations.
b) ... can be used to explain the actions of any system or organism.
c) ... cannot be improved because they depend on subjective interpretations.
d) ... entail that the ends always justify the means.

18. What are the main claims of anti-naturalist views of the social sciences? Select all correct alternatives.

a) The social sciences cannot and should not follow the methodology of the natural sciences.
b) Although the social sciences use qualitative data, in contrast to the natural sciences, there are methods by which these data can be quantified and thus be analyzed with natural science methods.
c) The social sciences are fundamentally different from the natural sciences, for instance because the social sciences work with qualitative data and case studies which the natural sciences do not.
d) The social sciences, in contrast to the natural sciences, are in the business of interpreting human behaviour. Interpretation does not produce causal explanations, but some other form of understanding.

19. A decision has to be made between four alternative courses of action.
On alternative A, there are two possible outcomes: 100 people dying (probability 0.6) or no one dying (probability 0.4).
On alternative B, there are four possible outcomes (equal probabilities): 20 people dying, 100 people dying, 400 people dying, or 4 people dying.
On C, there are two possible outcomes: 800 people dying (probability 0.01) or no one dying (probability 0.99).
On D, finally, there are two possible outcomes (equal probabilities): 30 people dying, or no one dying.
Select all correct statement(s), given the expected value criterion for decisions involving risks:

a) C is the second best alternative.
b) C is the best alternative.
c) A is the best alternative.
d) B is the fourth best alternative.

Downloaded by jonathan X (jonathantades@gmail.com)

20. Which of these are theories of normative ethics? Select all correct answers.

a) Informed consent.
b) Utilitarianism.
c) Moral justification.
d) Deontology.

Downloaded by jonathan X (jonathantades@gmail.com)

PART II – Account for and apply core course concepts

Instructions for Part II: Solve the following problems on separate sheets of paper. Be sure to apply the correct terminology. An answer is expected to be about one page – longer than a couple of sentences, but shorter than two full pages, depending on your handwriting.

1. **Explanations. (5 points)**

   One example of a causal explanation is a "belief-desire explanations". Explain what a "belief-desire explanation" is and how it can be used for interpretation.

   Use this form of explanation to interpret the following observation: ”When the first click on the cancel-button did not stop the process the subjects continued to click the cancel-button on average 10 additional times”.

   Another type of explanation is the deductive-nomological model of explanation. Account for what such an explanation is and how it works.

   Show how the deductive-nomological model of explanation could explain the following observation: ”The bullet passed through the bullet-resistant vest, rather than being stopped by it”.

   Finally, discuss one problem that deductive-nomological explanations might have, and properly explain and justify why this is a problem.

2. **Methodology and knowledge. (5 points)**

   First, explain the difference between scientific methods and scientific methodology.

   Then, give an example of a pragmatic justification and one example of an epistemic justification of the following case: "using a survey rather than a sleep lab to determine people's sleep quality".

   Now, account for the classical definition of knowledge by stating the criteria and describing what each criterion means.

   Then, consider this example: "Jonathan knows that vaccines cause headaches". Explain what is required for the statement to be true, according the classical definition of knowledge.

   Finally, discuss how scientific methodology can increase and improve scientific knowledge.

Downloaded by jonathan X (jonathantades@gmail.com)

PART III – Analyze an empirical study

Instructions: Read through the following case and then solve the problems below. As an indication, an answer a 4 point problem is expected to be about one page – longer than a couple of sentences, but shorter than two full pages. This, of course, depends on your handwriting, and there are no restrictions.

Case: Lack of sleep has been found to be connected to both poor health and lower productivity at work. A study investigated if nap-rooms at work could improve productivity. A large Swedish law-firm was selected for the test, given that lawyers are commonly believed to work very hard and long hours. Three conference rooms were refurnished with reclining comfortable chairs, and a speaker playing soothing music. Half of the personnel of each department in the law firm were randomly selected for the intervention, and the other half functioned as control.

The test ran for six days and was set up as follows. At 11:30 all employees were given a simple task to measure productivity. After lunch, the chosen employees were instructed to go to the nap-room and rest for 15-25 minutes. After 30 minutes these employees were woken up, if they had not woken up on their own, and then these employees were administered the same task again. To determine productivity the employees were asked to sort a large number of unsorted screws into their correct boxes (categorizing length, type, thickness, and distance between the threads on the screw) during 60 seconds. This task was chosen to give a clear quantitative measure, with low random error, and also took little time to perform. Only correctly sorted screws were considered.

The results showed big differences both inter-group and intra-group. Not only did productivity increase by 50 % post-nap of the final day of testing for the test group, but the test group also outperformed the control group by 70 % on the final day of testing. The researchers therefore concluded that implementing nap-rooms at companies and large organisations, could improve productivity with at least 50%.

Problem 1 (4 points). Discuss the construct validity of the operationalization of the feature of interest by: describing the operationalization in this case, evaluating the construct validity of this operationalization, discussing what other aspects of the feature of interest should have been measured, justifying why and giving examples of how this could have been done.

Problem 2 (4 points). Discuss what choices made in the experimental setup might have a negative impact on the internal validity by: pointing out flaws in the design of the study (beyond what problems there might be with the design of the operationalization), explaining how these are methodologically relevant, and by suggesting alternative ways of designing the test (i.e. improvements) that would not have a negative impact on the internal validity in this way.

Problem 3 (2 points). Structure the researchers' reasoning in the form of the hypothetico-deductive method. Your argument should have 3-6 premises and end with either rejecting or confirming the hypothesis "Nap-rooms cannot improve productivity". Then, explain what auxiliary assumptions (or auxiliary hypotheses) in this argument might be put into question by your solution of problems 1 and 2.

Downloaded by jonathan X (jonathantades@gmail.com)

Downloaded by jonathan X (jonathantades@gmail.com)

TaMoS: Answer sheet for mcq's

Student's name:

Personal number:

Mark correct options with "X".

	a)	b)	c)	d)
1.				
2.				
3.				
4.				
5.				
6.				
7.				
8.				
9.				
10.				

	a)	b)	c)	d)
11.				
12.				
13.				
14.				
15.				
16.				
17.				
18.				
19.				
20.				

To be filled out by corrector

Part I	Part II	Part III	Score	Bonus	Total	Grade	Initials

Downloaded by jonathan X (jonathantades@gmail.com)